Week 8: Database Relationship Analysis using Joins

Scenario

The e-commerce company stores customer, product, order, and payment information in separate tables. Joins combine related records to help management understand customer purchases, order details, and payment status.

INNER JOIN: Customer Details with Orders

Definition: INNER JOIN combines rows only when the join condition matches in both tables. Here, it connects customers to their orders so customer names, order IDs, and order amounts can be displayed together.

```
SQL> SELECT CONCAT(C.First_Name,C.Last_Name) AS Customer_Name,
  2  Orders.OrderID,
  3  Orders.TotalAmount
  4  FROM Customer C
  5  INNER JOIN Orders
  6  ON C.Customer_ID = Orders.CustomerID;

CUSTOMER_NAME                          ORDERID TOTALAMOUNT
-------------------------------------- ------- -----------
ArjunKumar                             ORD0001       55850
PriyaSharma                            ORD0002        3350
RahulMenon                             ORD0003       28000
AnanyaIyer                             ORD0004        8300
VikramReddy                            ORD0005       22000
MeeraKrishnan                          ORD0006        7000
SureshNair                             ORD0007        6500
KavyaRajan                             ORD0008        5700
KarthikSubramanian                     ORD0009       18500
DivyaPillai                            ORD0010        5200
AdityaVerma                            ORD0011        7000

CUSTOMER_NAME                          ORDERID TOTALAMOUNT
-------------------------------------- ------- -----------
SnehaPatel                             ORD0012        2450
LakshmiVenkatesan                      ORD0014       28000
NithinMurugan                          ORD0015        8350

14 rows selected.
```

INNER JOIN: Order Details with Payment Information

Definition: This query joins Orders and Payment using OrderID. It displays order dates and amounts alongside payment dates, payment methods, and payment statuses, but only for orders with matching payment records.

```
SQL> SELECT o.orderID,
  2  o.TotalAmount,
  3  o.OrderDate,
  4  o.CustomerID,
  5  p.PaymentDate,
  6  p.PaymentMode,
  7  p.PaymentStatus
  8  FROM Payment p JOIN Orders o
  9  ON p.OrderID = o.OrderID;

ORDERID TOTALAMOUNT ORDERDATE CUSTOMERID PAYMENTDA PAYMENTMODE      PAYMENTSTA
------- ----------- --------- ---------- --------- ---------------- ----------
ORD0001       55850 10-JAN-26 CUST000001 10-JAN-26 Credit Card      Completed
ORD0002        3350 25-JAN-26 CUST000002 25-JAN-26 UPI              Completed
ORD0004        8300 18-FEB-26 CUST000004 18-FEB-26 Debit Card       Completed
ORD0006        7000 15-MAR-26 CUST000006 15-MAR-26 Net Banking      Completed
ORD0008        5700 20-APR-26 CUST000008 20-APR-26 COD              Completed
ORD0010        5200 18-MAY-26 CUST000010 18-MAY-26 UPI              Completed
ORD0012        2450 20-JUN-26 CUST000012 20-JUN-26 Credit Card      Completed
ORD0014       28000 15-AUG-26 CUST000014 15-AUG-26 Debit Card       Completed
ORD0014       28000 15-AUG-26 CUST000014 14-AUG-26 Debit Card       Failed

9 rows selected.
```

INNER JOIN: Products Purchased by Customers

Definition: Joining Customer, Orders, OrderDetail, and Product links customers to the products they purchased. The result can show customer name, order ID, product, quantity, price, and total amount.

```
9 rows selected.

SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDERID AS ORDER_ID,
  5      P.PRODUCT_ID,
  6      P.PRODUCT_NAME,
  7      OD.QUANTITY,
  8      OD.PRICE,
  9      OD.QUANTITY * OD.PRICE AS TOTAL_AMOUNT
 10  FROM CUSTOMER C
 11  JOIN ORDERS O
 12      ON C.CUSTOMER_ID = O.CUSTOMERID
 13  JOIN ORDERDETAIL OD
 14      ON O.ORDERID = OD.ORDERID
 15  JOIN PRODUCT P
 16      ON OD.PRODUCTID = P.PRODUCT_ID
 17  ORDER BY C.CUSTOMER_ID, O.ORDERDATE;

CUSTOMER_I CUSTOMER_NAME                                ORDER_I PRODUCT
---------- -------------------------------------------- ------- -------
PRODUCT_NAME                                           QUANTITY      PRICE
------------------------------------------------------ ---------- ---------
TOTAL_AMOUNT
------------
CUST000001 Arjun Kumar                                  ORD0001 P000001
Laptop                                                         1      55000
       55000

CUST000001 Arjun Kumar                                  ORD0001 P000002
Wireless Mouse                                                 1        850
         850

CUSTOMER_I CUSTOMER_NAME                                ORDER_I PRODUCT
---------- -------------------------------------------- ------- -------
PRODUCT_NAME                                           QUANTITY      PRICE
------------------------------------------------------ ---------- ---------
TOTAL_AMOUNT
------------

CUST000002 Priya Sharma                                 ORD0002 P000006
USB Cable                                                      1        450
         450

CUST000002 Priya Sharma                                 ORD0002 P000004
Bluetooth Speaker                                              1       3200

CUSTOMER_I CUSTOMER_NAME                                ORDER_I PRODUCT
---------- -------------------------------------------- ------- -------
PRODUCT_NAME                                           QUANTITY      PRICE
------------------------------------------------------ ---------- ---------
TOTAL_AMOUNT
------------
        3200

CUST000003 Rahul Menon                                  ORD0003 P000005
Smartphone                                                     6      28000
      168000

CUST000004 Ananya Iyer                                  ORD0004 P000006
```

Product Purchase Results

Definition: This continuation displays additional rows from the multi-table purchase query. It demonstrates how joins combine customer, order, order-detail, and product records across the dataset.



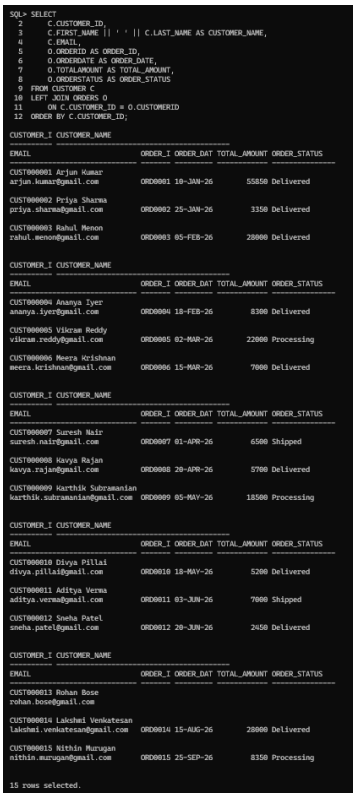
LEFT JOIN: All Customers and Their Orders

Definition: LEFT JOIN returns every row from the left table and matching rows from the right table. Customers with no matching order still appear, with NULL values for the order columns.



LEFT JOIN: Customers Who Have Not Placed Orders

Definition: A LEFT JOIN followed by a check for a NULL order identifier identifies customers with no matching purchase records. This is useful for finding customers who have not yet ordered.



```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      C.EMAIL,
  5      O.ORDERID AS ORDER_ID,
  6      O.ORDERDATE AS ORDER_DATE,
  7      O.TOTALAMOUNT AS TOTAL_AMOUNT,
  8      O.ORDERSTATUS AS ORDER_STATUS
  9  FROM CUSTOMER C
 10  LEFT JOIN ORDERS O
 11      ON C.CUSTOMER_ID = O.CUSTOMERID
 12  ORDER BY C.CUSTOMER_ID;

CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                          ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------------------ ------- --------- ------------ --------------
CUST000001 Arjun Kumar
arjun.kumar@gmail.com          ORD0001 10-JAN-26        55850 Delivered

CUST000002 Priya Sharma
priya.sharma@gmail.com         ORD0002 25-JAN-26         3350 Delivered

CUST000003 Rahul Menon
rahul.menon@gmail.com          ORD0003 05-FEB-26        28000 Delivered


CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                          ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------------------ ------- --------- ------------ --------------
CUST000004 Ananya Iyer
ananya.iyer@gmail.com          ORD0004 18-FEB-26         8300 Delivered

CUST000005 Vikram Reddy
vikram.reddy@gmail.com         ORD0005 02-MAR-26        22000 Processing

CUST000006 Meera Krishnan
meera.krishnan@gmail.com       ORD0006 15-MAR-26         7000 Delivered


CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                          ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------------------ ------- --------- ------------ --------------
CUST000007 Suresh Nair
suresh.nair@gmail.com          ORD0007 01-APR-26         6500 Shipped

CUST000008 Kavya Rajan
kavya.rajan@gmail.com          ORD0008 20-APR-26         5700 Delivered

CUST000009 Karthik Subramanian
karthik.subramanian@gmail.com  ORD0009 05-MAY-26        18500 Processing


CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                          ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------------------ ------- --------- ------------ --------------
CUST000010 Divya Pillai
divya.pillai@gmail.com         ORD0010 18-MAY-26         5200 Delivered

CUST000011 Aditya Verma
aditya.verma@gmail.com         ORD0011 03-JUN-26         7000 Shipped

CUST000012 Sneha Patel
sneha.patel@gmail.com          ORD0012 20-JUN-26         2450 Delivered


CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                          ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------------------ ------- --------- ------------ --------------
CUST000013 Rohan Bose
rohan.bose@gmail.com

CUST000014 Lakshmi Venkatesan
lakshmi.venkatesan@gmail.com   ORD0014 15-AUG-26        28000 Delivered

CUST000015 Nithin Murugan
nithin.murugan@gmail.com       ORD0015 25-SEP-26         8350 Processing

15 rows selected.
```

LEFT JOIN: Products with No Sales; RIGHT JOIN Introduction

Definition: A LEFT JOIN from Product to OrderDetail includes all products, even when no order-detail row exists. Checking for NULL order-detail product IDs identifies products with no recorded sales. RIGHT JOIN keeps all rows from its right-hand table and matching rows from the left.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      C.EMAIL,
  5      C.PHONE
  6  FROM CUSTOMER C
  7  LEFT JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  WHERE O.ORDERID IS NULL
 10  ORDER BY C.CUSTOMER_ID;

CUSTOMER_I CUSTOMER_NAME
---------- ------------------------------------------
EMAIL                                    PHONE
---------------------------------------- ----------
CUST000013 Rohan Bose
rohan.bose@gmail.com                     9654321098
```

RIGHT JOIN: Concept and Task

Definition: RIGHT JOIN returns all records from the right table and matching records from the left table. The following task uses it to display orders with customer information.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       P.PRICE
  5  FROM PRODUCT P
  6  LEFT JOIN ORDERDETAIL OD
  7       ON P.PRODUCT_ID = OD.PRODUCTID
  8  WHERE OD.PRODUCTID IS NULL
  9  ORDER BY P.PRODUCT_ID;

PRODUCT PRODUCT_NAME                                                 PRICE
------- ---------------------------------------------------------- ----------
P000007 Power Bank                                                   1800
P000011 Web Camera                                                   2800
```

RIGHT JOIN: Orders with Customer Information

Definition: RIGHT JOIN keeps all records from the right-hand table and includes matching customer details when available. It can reveal orders that lack a matching customer record.



```
SQL> SELECT
  2      O.ORDERID AS ORDER_ID,
  3      O.ORDERDATE AS ORDER_DATE,
  4      O.TOTALAMOUNT AS TOTAL_AMOUNT,
  5      O.ORDERSTATUS AS ORDER_STATUS,
  6      C.CUSTOMER_ID,
  7      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  8      C.EMAIL
  9  FROM CUSTOMER C
 10  RIGHT JOIN ORDERS O
 11      ON C.CUSTOMER_ID = O.CUSTOMERID
 12  ORDER BY O.ORDERDATE;

ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS    CUSTOMER_I
------- --------- ------------ --------------- ----------
CUSTOMER_NAME                                 EMAIL
--------------------------------------------- ----------------------------------------

ORD0001 10-JAN-26        55850 Delivered       CUST000001
Arjun Kumar                                   arjun.kumar@gmail.com

ORD0002 25-JAN-26         3350 Delivered       CUST000002
Priya Sharma                                  priya.sharma@gmail.com

ORD0003 05-FEB-26        28000 Delivered       CUST000003
Rahul Menon                                   rahul.menon@gmail.com


ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS    CUSTOMER_I
------- --------- ------------ --------------- ----------
CUSTOMER_NAME                                 EMAIL
--------------------------------------------- ----------------------------------------

ORD0004 18-FEB-26         8300 Delivered       CUST000004
Ananya Iyer                                   ananya.iyer@gmail.com

ORD0005 02-MAR-26        22000 Processing      CUST000005
Vikram Reddy                                  vikram.reddy@gmail.com

ORD0006 15-MAR-26         7000 Delivered       CUST000006
Meera Krishnan                                meera.krishnan@gmail.com


ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS    CUSTOMER_I
------- --------- ------------ --------------- ----------
CUSTOMER_NAME                                 EMAIL
--------------------------------------------- ----------------------------------------

ORD0007 01-APR-26         6500 Shipped         CUST000007
Suresh Nair                                   suresh.nair@gmail.com

ORD0008 20-APR-26         5700 Delivered       CUST000008
Kavya Rajan                                   kavya.rajan@gmail.com

ORD0009 05-MAY-26        18500 Processing      CUST000009
Karthik Subramanian                           karthik.subramanian@gmail.com


ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS    CUSTOMER_I
------- --------- ------------ --------------- ----------
CUSTOMER_NAME                                 EMAIL
--------------------------------------------- ----------------------------------------

ORD0010 18-MAY-26         5200 Delivered       CUST000010
Divya Pillai                                  divya.pillai@gmail.com

ORD0011 03-JUN-26         7000 Shipped         CUST000011
Aditya Verma                                  aditya.verma@gmail.com

ORD0012 20-JUN-26         2450 Delivered       CUST000012
Sneha Patel                                   sneha.patel@gmail.com


ORDER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS    CUSTOMER_I
------- --------- ------------ --------------- ----------
CUSTOMER_NAME                                 EMAIL
--------------------------------------------- ----------------------------------------

ORD0014 15-AUG-26        28000 Delivered       CUST000014
Lakshmi Venkatesan                            lakshmi.venkatesan@gmail.com

ORD0015 25-SEP-26         8350 Processing      CUST000015
Nithin Murugan                                nithin.murugan@gmail.com


14 rows selected.
```

RIGHT JOIN: Orders with Missing Customer Details

Definition: Checking for NULL customer identifiers after joining orders and customers identifies orders whose customer details are missing.

```
SQL> SELECT
  2       O.ORDERID AS ORDER_ID,
  3       O.ORDERDATE AS ORDER_DATE,
  4       O.TOTALAMOUNT AS TOTAL_AMOUNT,
  5       O.ORDERSTATUS AS ORDER_STATUS
  6  FROM CUSTOMER C
  7  RIGHT JOIN ORDERS O
  8       ON C.CUSTOMER_ID = O.CUSTOMERID
  9  WHERE C.CUSTOMER_ID IS NULL;

no rows selected
```

RIGHT JOIN: Payment Records with Order Details

Definition: This query combines payment and order information so every payment record can be viewed with its associated order details. It supports checking payment mode, status, transaction amount, and order status.

```
SQL> SELECT
  2      P.PAYMENT_ID,
  3      P.PAYMENTDATE AS PAYMENT_DATE,
  4      P.PAYMENTMODE AS PAYMENT_MODE,
  5      P.PAYMENTSTATUS AS PAYMENT_STATUS,
  6      P.TRANSACTIONAMOUNT AS TRANSACTION_AMOUNT,
  7      O.ORDERID AS ORDER_ID,
  8      O.ORDERDATE AS ORDER_DATE,
  9      O.TOTALAMOUNT AS ORDER_AMOUNT,
 10      O.ORDERSTATUS AS ORDER_STATUS
 11  FROM ORDERS O
 12  RIGHT JOIN PAYMENT P
 13      ON O.ORDERID = P.ORDERID
 14  ORDER BY P.PAYMENTDATE;

PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT ORDER_I
------- --------- --------------- ---------- ------------------ ------
ORDER_DAT ORDER_AMOUNT ORDER_STATUS
--------- ------------ ---------------
PAY0001 10-JAN-26 Credit Card     Completed              55850 ORD0001
10-JAN-26         55850 Delivered

PAY0002 25-JAN-26 UPI             Completed               3350 ORD0002
25-JAN-26          3350 Delivered

PAY0003 18-FEB-26 Debit Card      Completed               8300 ORD0004
18-FEB-26          8300 Delivered


PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT ORDER_I
------- --------- --------------- ---------- ------------------ ------
ORDER_DAT ORDER_AMOUNT ORDER_STATUS
--------- ------------ ---------------
PAY0004 15-MAR-26 Net Banking     Completed               7000 ORD0006
15-MAR-26          7000 Delivered

PAY0005 20-APR-26 COD             Completed               5700 ORD0008
20-APR-26          5700 Delivered

PAY0006 18-MAY-26 UPI             Completed               5200 ORD0010
18-MAY-26          5200 Delivered


PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT ORDER_I
------- --------- --------------- ---------- ------------------ ------
ORDER_DAT ORDER_AMOUNT ORDER_STATUS
--------- ------------ ---------------
PAY0007 20-JUN-26 Credit Card     Completed               2450 ORD0012
20-JUN-26          2450 Delivered

PAY0009 14-AUG-26 Debit Card      Failed                 28000 ORD0014
15-AUG-26         28000 Delivered

PAY0008 15-AUG-26 Debit Card      Completed              28000 ORD0014
15-AUG-26         28000 Delivered


9 rows selected.
```

Complete Order Details Report

Definition: A complete order report joins customer, order, order-detail, product, and payment-related information to show customer name, product name, quantity, order date, total amount, and payment status.



```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      P.PRODUCT_NAME,
  4      OD.QUANTITY,
  5      O.ORDERDATE AS ORDER_DATE,
  6      O.TOTALAMOUNT AS TOTAL_AMOUNT,
  7      PM.PAYMENTSTATUS AS PAYMENT_STATUS
  8  FROM CUSTOMER C
  9  JOIN ORDERS O
 10      ON C.CUSTOMER_ID = O.CUSTOMERID
 11  JOIN ORDERDETAIL OD
 12      ON O.ORDERID = OD.ORDERID
 13  JOIN PRODUCT P
 14      ON OD.PRODUCTID = P.PRODUCT_ID
 15  LEFT JOIN PAYMENT PM
 16      ON O.ORDERID = PM.ORDERID
 17  ORDER BY O.ORDERDATE, C.CUSTOMER_ID;

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
Arjun Kumar
Laptop                                                    1 10-JAN-26
       55850 Completed

Arjun Kumar
Wireless Mouse                                            1 10-JAN-26
       55850 Completed

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------

Priya Sharma
USB Cable                                                 1 25-JAN-26
        3350 Completed

Priya Sharma
Bluetooth Speaker                                         1 25-JAN-26

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
        3350 Completed

Rahul Menon
Smartphone                                                6 05-FEB-26
       28000

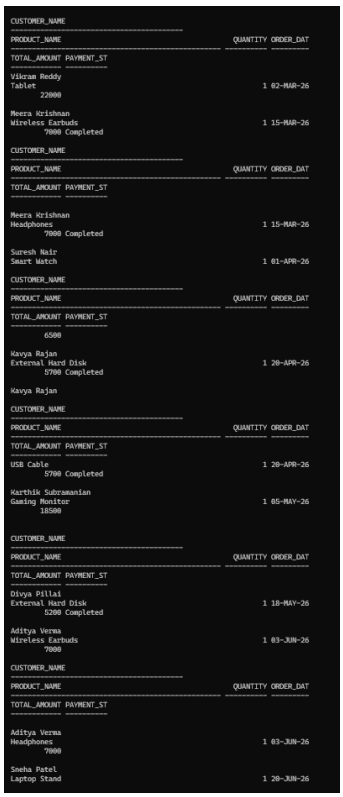
Ananya Iyer

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
USB Cable                                                 4 18-FEB-26
        8300 Completed

Ananya Iyer
Smart Watch                                               1 18-FEB-26
        8300 Completed
```

Complete Order Details: Continued Results

Definition: These rows continue the complete order report, showing how a single customer or order can be associated with one or more purchased products and their quantities and amounts.



```
CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
Vikram Reddy
Tablet                                                    1 02-MAR-26
       22000

Meera Krishnan
Wireless Earbuds                                          1 15-MAR-26
        7000 Completed

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------

Meera Krishnan
Headphones                                                1 15-MAR-26
        7000 Completed

Suresh Nair
Smart Watch                                               1 01-APR-26

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
        6500

Kavya Rajan
External Hard Disk                                        1 20-APR-26
        5700 Completed

Kavya Rajan

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
USB Cable                                                 1 20-APR-26
        5700 Completed

Karthik Subramanian
Gaming Monitor                                            1 05-MAY-26
       18500


CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
Divya Pillai
External Hard Disk                                        1 18-MAY-26
        5200 Completed

Aditya Verma
Wireless Earbuds                                          1 03-JUN-26
        7000

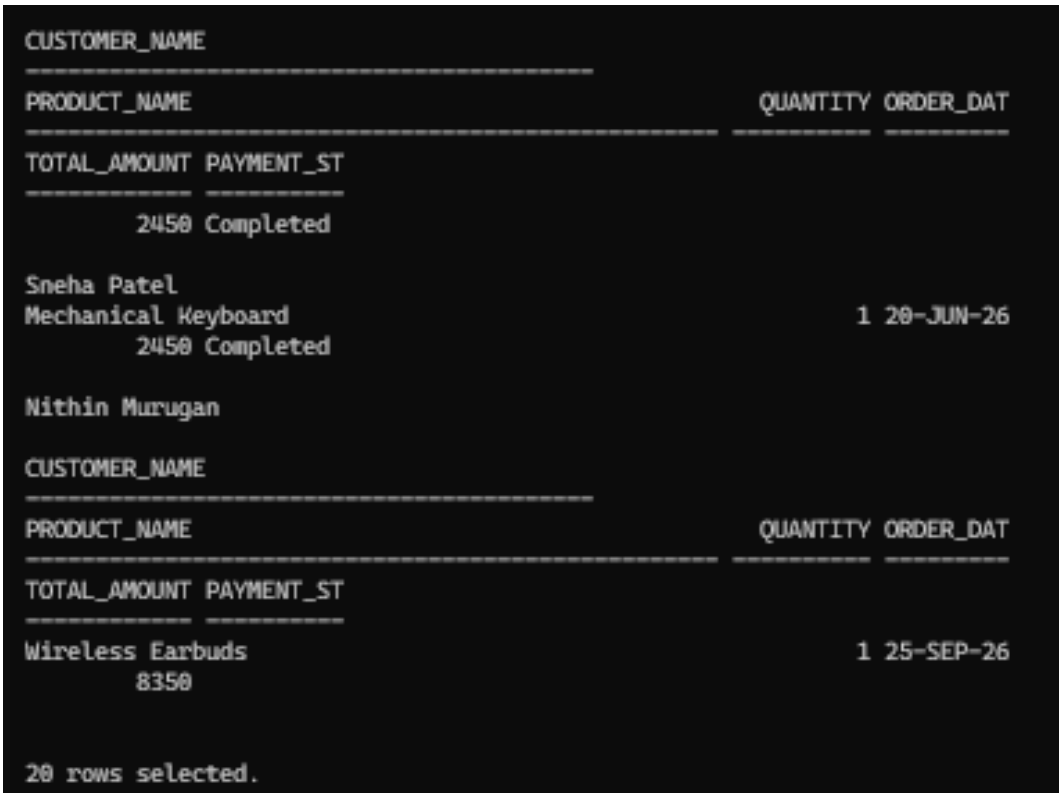
CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------

Aditya Verma
Headphones                                                1 03-JUN-26
        7000

Sneha Patel
Laptop Stand                                              1 20-JUN-26
```

Complete Order Details: Further Results

Definition: This continuation presents additional joined order records and illustrates the relationship between customer, product, quantity, order date, amount, and payment status.



```
CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
        2450 Completed

Sneha Patel
Mechanical Keyboard                                       1 20-JUN-26
        2450 Completed

Nithin Murugan

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       QUANTITY ORDER_DAT
-------------------------------------------------- -------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
Wireless Earbuds                                          1 25-SEP-26
        8350


20 rows selected.
```

Customer Purchase History: Products Purchased

Definition: Customer purchase history joins orders and order details to products. Filtering by a customer ID lists the products that customer purchased, along with the relevant order date and quantity.



```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      O.ORDERID AS ORDER_ID,
  4      O.ORDERDATE AS ORDER_DATE,
  5      P.PRODUCT_NAME,
  6      OD.QUANTITY,
  7      OD.PRICE
  8  FROM CUSTOMER C
  9  JOIN ORDERS O
 10      ON C.CUSTOMER_ID = O.CUSTOMERID
 11  JOIN ORDERDETAIL OD
 12      ON O.ORDERID = OD.ORDERID
 13  JOIN PRODUCT P
 14      ON OD.PRODUCTID = P.PRODUCT_ID
 15  WHERE C.CUSTOMER_ID = 'CUST000006'
 16  ORDER BY O.ORDERDATE;

CUSTOMER_NAME                             ORDER_I ORDER_DAT
----------------------------------------- ------- ---------
PRODUCT_NAME                                        QUANTITY      PRICE
--------------------------------------------------- ---------- ----------
Meera Krishnan                            ORD0006 15-MAR-26
Wireless Earbuds                                         1       3500

Meera Krishnan                            ORD0006 15-MAR-26
Headphones                                               1       4200
```

Customer Purchase History: Total Amount Spent

Definition: SUM groups the order amounts by customer to calculate total spending and rank customers by their purchase value.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      ROUND(SUM(O.TOTALAMOUNT), 2) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_I CUSTOMER_NAME                              TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000001 Arjun Kumar                                             55850
CUST000014 Lakshmi Venkatesan                                      28000
CUST000003 Rahul Menon                                             28000
CUST000005 Vikram Reddy                                            22000
CUST000009 Karthik Subramanian                                     18500
CUST000015 Nithin Murugan                                           8350
CUST000004 Ananya Iyer                                              8300
CUST000006 Meera Krishnan                                           7000
CUST000011 Aditya Verma                                             7000
CUST000007 Suresh Nair                                              6500
CUST000008 Kavya Rajan                                              5700

CUSTOMER_I CUSTOMER_NAME                              TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000010 Divya Pillai                                             5200
CUST000002 Priya Sharma                                             3350
CUST000012 Sneha Patel                                              2450

14 rows selected.
```

Customer Purchase History: Total Amount Spent

Definition: SUM calculates the total amount spent by each customer. GROUP BY creates one summary per customer, and sorting the totals in descending order highlights the highest spenders.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY NUMBER_OF_ORDERS DESC;

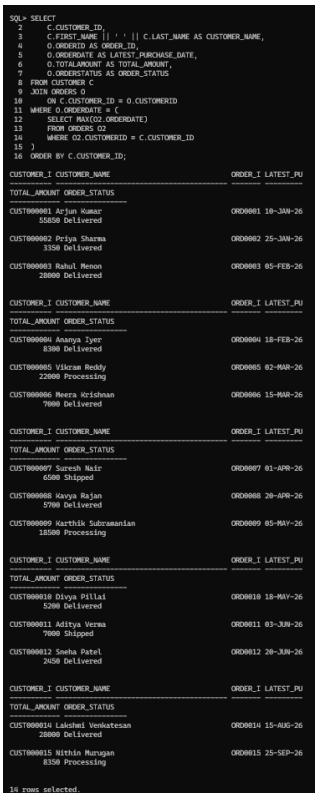
CUSTOMER_I CUSTOMER_NAME                              NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000001 Arjun Kumar                                              1
CUST000015 Nithin Murugan                                           1
CUST000003 Rahul Menon                                              1
CUST000004 Ananya Iyer                                              1
CUST000005 Vikram Reddy                                             1
CUST000006 Meera Krishnan                                           1
CUST000007 Suresh Nair                                              1
CUST000008 Kavya Rajan                                              1
CUST000009 Karthik Subramanian                                      1
CUST000010 Divya Pillai                                             1
CUST000011 Aditya Verma                                             1

CUSTOMER_I CUSTOMER_NAME                              NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000012 Sneha Patel                                              1
CUST000002 Priya Sharma                                             1
CUST000014 Lakshmi Venkatesan                                       1
CUST000013 Rohan Bose                                               0

15 rows selected.
```

Customer Purchase History: Number of Orders

Definition: COUNT counts order records for each customer. LEFT JOIN also keeps customers who have no orders, allowing their order count to appear as zero or be handled with a suitable count expression.



```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDERID AS ORDER_ID,
  5      O.ORDERDATE AS LATEST_PURCHASE_DATE,
  6      O.TOTALAMOUNT AS TOTAL_AMOUNT,
  7      O.ORDERSTATUS AS ORDER_STATUS
  8  FROM CUSTOMER C
  9  JOIN ORDERS O
 10      ON C.CUSTOMER_ID = O.CUSTOMERID
 11  WHERE O.ORDERDATE = (
 12      SELECT MAX(O2.ORDERDATE)
 13      FROM ORDERS O2
 14      WHERE O2.CUSTOMERID = C.CUSTOMER_ID
 15  )
 16  ORDER BY C.CUSTOMER_ID;

CUSTOMER_I CUSTOMER_NAME                                    ORDER_I LATEST_PU
---------- -------------------------------------------------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ---------------
CUST000001 Arjun Kumar                                      ORD0001 10-JAN-26
       55850 Delivered

CUST000002 Priya Sharma                                     ORD0002 25-JAN-26
        3350 Delivered

CUST000003 Rahul Menon                                      ORD0003 05-FEB-26
       28000 Delivered


CUSTOMER_I CUSTOMER_NAME                                    ORDER_I LATEST_PU
---------- -------------------------------------------------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ---------------
CUST000004 Ananya Iyer                                      ORD0004 18-FEB-26
        8300 Delivered

CUST000005 Vikram Reddy                                     ORD0005 02-MAR-26
       22000 Processing

CUST000006 Meera Krishnan                                   ORD0006 15-MAR-26
        7000 Delivered


CUSTOMER_I CUSTOMER_NAME                                    ORDER_I LATEST_PU
---------- -------------------------------------------------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ---------------
CUST000007 Suresh Nair                                      ORD0007 01-APR-26
        6500 Shipped

CUST000008 Kavya Rajan                                      ORD0008 20-APR-26
        5700 Delivered

CUST000009 Karthik Subramanian                              ORD0009 05-MAY-26
       18500 Processing


CUSTOMER_I CUSTOMER_NAME                                    ORDER_I LATEST_PU
---------- -------------------------------------------------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ---------------
CUST000010 Divya Pillai                                     ORD0010 18-MAY-26
        5200 Delivered

CUST000011 Aditya Verma                                     ORD0011 03-JUN-26
        7000 Shipped

CUST000012 Sneha Patel                                      ORD0012 20-JUN-26
        2450 Delivered


CUSTOMER_I CUSTOMER_NAME                                    ORDER_I LATEST_PU
---------- -------------------------------------------------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ---------------
CUST000014 Lakshmi Venkatesan                               ORD0014 15-AUG-26
       28000 Delivered

CUST000015 Nithin Murugan                                   ORD0015 25-SEP-26
        8350 Processing

14 rows selected.
```

Customer Purchase History: Latest Purchase Details

Definition: A latest-purchase query identifies the most recent order for each customer by comparing order dates. The output can include customer name, order ID, date, amount, and status.

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      O.ORDERID AS ORDER_ID,
  4      O.ORDERDATE AS ORDER_DATE,
  5      O.ORDERSTATUS AS ORDER_STATUS
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  ORDER BY O.ORDERDATE;

CUSTOMER_NAME                              ORDER_I ORDER_DAT ORDER_STATUS
------------------------------------------ ------- --------- ---------------
Arjun Kumar                                ORD0001 10-JAN-26 Delivered
Priya Sharma                               ORD0002 25-JAN-26 Delivered
Rahul Menon                                ORD0003 05-FEB-26 Delivered
Ananya Iyer                                ORD0004 18-FEB-26 Delivered
Vikram Reddy                               ORD0005 02-MAR-26 Processing
Meera Krishnan                             ORD0006 15-MAR-26 Delivered
Suresh Nair                                ORD0007 01-APR-26 Shipped
Kavya Rajan                                ORD0008 20-APR-26 Delivered
Karthik Subramanian                        ORD0009 05-MAY-26 Processing
Divya Pillai                               ORD0010 18-MAY-26 Delivered
Aditya Verma                               ORD0011 03-JUN-26 Shipped

CUSTOMER_NAME                              ORDER_I ORDER_DAT ORDER_STATUS
------------------------------------------ ------- --------- ---------------
Sneha Patel                                ORD0012 20-JUN-26 Delivered
Lakshmi Venkatesan                         ORD0014 15-AUG-26 Delivered
Nithin Murugan                             ORD0015 25-SEP-26 Processing

14 rows selected.
```

Sales Report: Quantity Sold and Total Revenue

Definition: The sales report groups order-detail rows by product. SUM of quantities gives quantity sold, while quantity multiplied by unit price contributes to total revenue.

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_NAME                                       QUANTITY_SOLD TOTAL_REVENUE
-------------------------------------------------- ------------- -------------
Smartphone                                                     6        168000
Laptop                                                         1         55000
Tablet                                                         1         22000
Gaming Monitor                                                 1         18500
Smart Watch                                                    2         13000
Wireless Earbuds                                               3         10500
External Hard Disk                                             2         10400
Headphones                                                     2          8400
Bluetooth Speaker                                              1          3200
USB Cable                                                      6          2700
Mechanical Keyboard                                            1          2500

PRODUCT_NAME                                       QUANTITY_SOLD TOTAL_REVENUE
-------------------------------------------------- ------------- -------------
Laptop Stand                                                   1          1500
Wireless Mouse                                                 1           850

13 rows selected.
```

Business Reports: Customer Order Report and Sales Report

Definition: The Customer Order Report lists customer name, order ID, date, and status. The Sales Report groups order details by product to calculate quantity sold and total revenue.

```
SQL> SELECT
  2      PAYMENTMODE AS PAYMENT_MODE,
  3      COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  4      SUM(
  5          CASE
  6              WHEN PAYMENTSTATUS = 'Completed' THEN 1
  7              ELSE 0
  8          END
  9      ) AS SUCCESSFUL_PAYMENTS
 10  FROM PAYMENT
 11  GROUP BY PAYMENTMODE
 12  ORDER BY NUMBER_OF_TRANSACTIONS DESC;

PAYMENT_MODE    NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------- ---------------------- --------------------
Debit Card                           3                    2
UPI                                  2                    2
Credit Card                          2                    2
COD                                  1                    1
Net Banking                          1                    1
```

Customer Purchase Analysis: Top Purchasing Customers

Definition: This report orders customers by their total spending to identify the customers contributing the greatest purchase value.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                              TOTAL_SPENDING
---------- ---------------------------------------- --------------
CUST000001 Arjun Kumar                                        55850
CUST000014 Lakshmi Venkatesan                                 28000
CUST000003 Rahul Menon                                        28000
CUST000005 Vikram Reddy                                       22000
CUST000009 Karthik Subramanian                                18500
CUST000015 Nithin Murugan                                      8350
CUST000004 Ananya Iyer                                         8300
CUST000006 Meera Krishnan                                      7000
CUST000011 Aditya Verma                                        7000
CUST000007 Suresh Nair                                         6500
CUST000008 Kavya Rajan                                         5700

CUSTOMER_I CUSTOMER_NAME                              TOTAL_SPENDING
---------- ---------------------------------------- --------------
CUST000010 Divya Pillai                                        5200
CUST000002 Priya Sharma                                        3350
CUST000012 Sneha Patel                                         2450

14 rows selected.
```

Payment Analysis and Customer Purchase Analysis

Definition: The payment report groups records by payment mode, counts transactions, and counts successful payments. Customer purchase analysis compares total spending, order counts, and customer purchasing activity.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) = (
 13      SELECT MAX(ORDER_COUNT)
 14      FROM (
 15          SELECT COUNT(*) AS ORDER_COUNT
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  );

CUSTOMER_I CUSTOMER_NAME                              NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000001 Arjun Kumar                                              1
CUST000002 Priya Sharma                                             1
CUST000003 Rahul Menon                                              1
CUST000004 Ananya Iyer                                              1
CUST000005 Vikram Reddy                                             1
CUST000006 Meera Krishnan                                           1
CUST000007 Suresh Nair                                              1
CUST000008 Kavya Rajan                                              1
CUST000009 Karthik Subramanian                                      1
CUST000010 Divya Pillai                                             1
CUST000011 Aditya Verma                                             1

CUSTOMER_I CUSTOMER_NAME                              NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000012 Sneha Patel                                              1
CUST000014 Lakshmi Venkatesan                                       1
CUST000015 Nithin Murugan                                           1

14 rows selected.
```

Customer Purchase Analysis: Highest Spending

Definition: This result identifies customers whose total spending equals the maximum total spending calculated across customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTALAMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTALAMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  );

CUSTOMER_I CUSTOMER_NAME                              TOTAL_SPENDING
---------- ---------------------------------------- --------------
CUST000001 Arjun Kumar                                       55850
```